

DSA / Algorithms (Abdul Bari) - Day 40

45-day plan mapped to the Algorithms playlist

0/1 Knapsack (Branch & Bound)

0/1 knapsack solved with branch and bound.

Watch these Abdul Bari videos

> 7.2 0/1 Knapsack using Branch and Bound

Key concepts

- Compute upper bounds to prune.
- Explore include/exclude tree.
- Often faster than brute force.

Key idea

Bound = current value + fractional fill of the rest.

Tasks (1.5h)

- ☐ Watch the video
- ☐ Compute a node bound
- ☐ Solve a small instance

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0